

7	(a) sketch: concentric circles (<i>minimum of 3 circles</i>) separation increasing with distance from wire correct direction	M1 A1 B1	[3]
	(b) (i) arrow direction from wire B towards wire A	B1	[1]
	(ii) <i>either</i> reference to Newton's third law <i>or</i> force on each wire proportional to product of the two currents so forces are equal	M1 A1	[2]
	(c) force <u>always</u> towards wire A/ <u>always</u> in same direction varies from zero (to a maximum value) (1) variation is sinusoidal / $\sin^2$ (1) (at) twice frequency of current (1) (<i>any two, one each</i>)	B1 B2	[3]
8	(a) packet/quantum/discrete amount of energy	M1	